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Soft tissue fixation system

Abstract

A single delivery needle configuration accommodates multiple soft tissue fixation devices at once. The configuration eliminates the need to reload a fixation delivery device when multiple fixation devices are required. Further, this configuration can include different variations of fixation devices. In embodiments, these fixation devices are typically manufactured from standard braided polyester suture that is cut into segments with predetermined lengths. This design allows the braided suture segments to be preloaded with a suture threaded through the preloaded braided suture segments with a slip knot placed on the proximal end of a first braided suture segments (e.g., distal fixation device). The tailing suture from the slip knot passes through the second, third, fourth braided suture segments, or any required quantity of braided suture segments that have been preloaded into the delivery device. The suture passes completely through a delivery device.

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Background/Summary

PRIORITY (1) This patent application claims priority from provisional U.S. patent application No. 63/319,467, filed Mar. 14, 2022, entitled, “SOFT TISSUE FIXATION SYSTEM,” and naming Rickey Hart, Brian Tinkham, and Peter Rogal as inventors, the disclosure of which is incorporated herein, in its entirety, by reference.

FIELD

(1) Illustrative embodiments of the invention generally relate to surgical tissue fixation and, more

particularly, various embodiments of the invention relate to devices and methods of deploying the devices to fixate tissue.

BACKGROUND

(2) Surgical Fixation devices have primarily been manufactured from stainless, titanium, non-absorbable plastic, absorbable plastic or some hybrid of plastic and ceramic. However, the use of flexible fixation devices is far less common.

(3) There is a particular need in medical treatment settings of fixating soft tissues, such as closing gastroenterological defects, and for that matter devices that could be used for any soft tissue to tissue approximation.

SUMMARY OF VARIOUS EMBODIMENTS

(4) In accordance with one embodiment of the invention, a flexible braided anchor assembly for tissue fixation includes a flexible strand (e.g., suture), and a plurality of braided anchor segments slidably coupled to the flexible strand. The flexible strand passes through a bore hole that traverses from a proximal end to a distal end of each of the plurality of braided segments. The proximal end of a first braided anchor segment is adjacent to a distal end of the second braided anchor segment. The plurality of braided anchor segments are cinched together forming a flexible braided anchor assembly to provide tissue fixation.

(5) The plurality of braided anchor segments may include straight braided suture segments that have been deformed by the application of tension on the flexible strand. The flexible strand may be a monofilament. The flexible strand may be a suture. The suture may be a braided polyester suture. Each of the plurality of braided anchor segments may be a standard braided polyester suture that is cut into predetermined lengths.

(6) Each of the plurality of braided suture segments slidably coupled to the flexible strand may be treated to increase a rigidity of the plurality of braided suture segments slidably coupled to the flexible strand. Such a treatment may include heating each of the plurality of braided suture segments. Such a treatment may also include applying a cyanoacrylate coating to each of the plurality of braided suture segments.

(7) The cinched plurality of braided anchor segments may have a purse string effect on the plurality of braided anchor segments such that the plurality of flexible braided anchor segments is gathered to provide tissue fixation.

(8) In accordance with another embodiment of the invention, a method for implanting a plurality of braided suture segments connected via a single flexible strand for tissue fixation includes puncturing the tissue at a first location with a delivery needle and extruding a first braided suture segment from the delivery needle. The first braided suture segment is the first of a plurality of braided suture segments slidably coupled to a flexible strand. The flexible strand is threaded through a bore hole in the first braided suture segment.

(9) The method further includes pulling the flexible strand a first time to deform the first braided suture segment forming the first braided anchor segment. Pulling the flexible strand the first time tightens a slipknot around a proximal end of the first braided suture segment, and tightening the slipknot is configured to anchor the first braided anchor segment to the tissue to prevent the first braided anchor segment from pulling thru the tissue.

(10) The method further includes withdrawing the delivery needle from the tissue a first time, puncturing the tissue at a second location with the delivery needle, extruding a second braided suture segment from the delivery needle, and pulling the suture a second time to deform the second braided suture segment forming the second braided anchor segment.

(11) The method further includes withdrawing the delivery needle from the tissue a second time and pulling the suture a third time. The pulled suture causes the second anchor segment to be moved toward the first braided anchor segment.

(12) The method further includes securing the suture to keep the tissue fixated by the plurality of braided suture segments, and cutting the suture. The suture is cut a distance away from the

proximal end of a final braided anchor segment of the plurality of the braided anchor segments.

(13) Following pulling the suture a third time, and before securing the suture, the method may further include puncturing the tissue at a third location with the delivery needle. The method may further include extruding a third braided suture segment from the delivery needle. The method may further include pulling the suture a fourth time to deform the third braided suture segment forming the third braided anchor segment. The method may further include withdrawing the delivery needle from the tissue a third time. The method may further include pulling the suture a fourth time, and the pulled suture may cause the third braided anchor segment to be moved toward the second braided anchor segment.

(14) The first braided suture segment may be a primary first braided suture segment. Extruding the first braided suture segment may be accomplished by advancing first drive tube to push the first braided suture segment out of the delivery needle. After advancing the first drive tube to push the first braided suture segment out of the delivery needle, the first drive tube remains in an advanced position temporarily.

(15) Extruding a second braided suture segment may be accomplished by advancing second drive tube to push the second braided suture segment out of the delivery needle. The second location may be between about 5 mm and about 20 mm from the first location.

(16) Securing the suture may include a locking device or an overhead knot that can be run down to the fixation site. Securing the suture may include locking the suture in place using a locking mechanism comprising at least one of a tag, a barb, a cinch, a toggled needle, or a distal knot

(17) In accordance with yet another embodiment of the invention, a surgical device for implanting a flexible braided anchor assembly for tissue fixation includes a handle assembly. The handle assembly has a cylindrical bore passing through the handle assembly on a center axis along a length of the handle assembly.

(18) The surgical device further includes an outer tube positioned in the cylindrical bore of the handle assembly. The outer tube is hollow and has a common center line with the cylindrical bore of the handle assembly. The outer tube includes a delivery needle having a needle tip at a distal end of the outer tube. The needle tip is a sharp point on the delivery needle.

(19) The surgical device further includes a plurality of braided suture segments slidably coupled to a flexible strand, the flexible strand being translationally threaded through plurality of braided suture segments.

(20) The surgical device further includes a first drive tube positioned in the outer tube and having a common center line with the outer tube. The first drive tube is configured to extrude a first braided suture segment.

(21) The surgical device further includes a second drive tube positioned in the first drive tube and having a common center line with the first drive tube. The second drive tube is configured to extrude the remaining braided suture segments of the plurality of braided suture segments. The remaining braided suture segments of the plurality of braided suture segments are translationally threaded through and pressure fit within the first drive tube. Each of the plurality of braided suture segments may have a hollow braided suture segment.

(22) The suture may be a monofilament strand. The plurality of braided suture segments may include a plurality of hollow braided suture segments aligned end to end having the monofilament strand threaded through the aligned plurality of hollow braided suture segments.

(23) A column strength of the plurality of hollow braided suture segments may be increased by application of at least one of heat or a cyanoacrylate coating to each of the segments of the plurality of hollow braided suture segments.

(24) A braided suture segment of the plurality of braided suture segments may have a length of between about 5 mm and about 50 mm. The braided suture segment of the plurality of braided suture segments may have a length of between about 5 mm and about 12 mm.

(25) A distal end of the flexible strand threaded through the first braided suture segment may be

looped around the first braided suture segment and knotted around the flexible strand between the first braided suture segment and the first drive tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Those skilled in the art should more fully appreciate advantages of various embodiments of the invention from the following “Description of Illustrative Embodiments,” discussed with reference to the drawings summarized immediately below.
- (2) FIG. 1A shows a schematic plan view of a flexible braided anchor assembly for tissue fixation according to one embodiment of the present disclosure before cinching the suture tight.
- (3) FIG. 1B shows a schematic plan view of a flexible braided anchor assembly for tissue fixation according to one embodiment of the present disclosure after cinching the suture tight.
- (4) FIG. 2A shows a schematic cross section illustration of a delivery device with a needle after puncturing tissue according to an embodiment of the present disclosure.
- (5) FIG. 2B shows a schematic cross section illustration of a delivery device with a first braided suture segment after being extruded from the needle on the distal side of the tissue.
- (6) FIG. 2C shows a schematic cross section illustration of a delivery device with a first braided anchor segment after the suture has been pulled taut a first time.
- (7) FIG. 2D shows a schematic cross section illustration of a delivery device with the needle being repositioned above the tissue prior to puncturing the tissue in a second location.
- (8) FIG. 2E shows a schematic cross section illustration of a delivery device with a second braided suture segment partially extruded from the needle.
- (9) FIG. 2F shows a schematic cross section illustration of a delivery device with a second braided anchor segment after the suture has been pulled taut a second time.
- (10) FIG. 2G shows a schematic cross section illustration of a delivery device with the first braided anchor segment and second braided anchor segment after the needle has been withdrawn from the tissue and suture has been pulled taut a third time.
- (11) FIG. 2H shows a schematic cross section illustration of a delivery device with the first braided anchor segment, second braided anchor segment, and a third braided anchor segment.
- (12) FIG. 3 is a cross section of an embodiment of a delivery device with portion of a delivery device including an outer tube having a delivery needle with a needle tip.
- (13) FIG. 4 is a cross section of an embodiment of a delivery device with a portion of a delivery device that has punctured the tissue and has delivered the first braided suture segment to a distal side of tissue.
- (14) FIG. 5A shows an embodiment of a braided suture segment.
- (15) FIG. 5B shows another embodiment of a braided suture segment.
- (16) FIG. 5C shows another embodiment of a braided suture segment.
- (17) FIG. 6A shows yet another embodiment of a braided suture segment.
- (18) FIG. 6B shows yet another embodiment of a braided suture segment.
- (19) FIG. 7A shows yet another embodiment of a braided suture segment.
- (20) FIG. 7B shows yet another embodiment of a braided suture segment.
- (21) FIG. 8 shows still another embodiment of a braided suture segment.
- (22) FIG. 9A shows an embodiment of a delivery device with a handle assembly with handle cover screwed to handle base.
- (23) FIG. 9B shows an embodiment of a delivery device with a handle assembly with the handle cover removed for visualization.
- (24) FIG. 10 shows a schematic cross section of an embodiment of a delivery device.
- (25) FIG. 11 shows embodiments of a delivery device with an outer tube, a first drive tube, a second drive tube, and an assembly of braided suture segments strung on a flexible filament.

(25) FIG. 12 shows steps of an embodiment of a method for implanting braided anchor segments connected via a single flexible strand for tissue fixation.

DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(26) The present disclosure includes a single delivery needle configuration which can accompany multiple soft tissue fixation devices at once. The device addresses tissue to tissue fixation. The configuration eliminates the need to reload a fixation delivery device when multiple fixation devices are required. Further, this configuration can include different variations of devices via: length; diameter; connectivity; function; and deployment. In embodiments, these presently disclosed fixation devices are typically manufactured from standard braided polyester suture that is cut into predetermined lengths.

(27) This design would be preloaded with either a monofilament or a braided type suture with a slip knot placed on the proximal end of the distal fixation device. The tailing suture from the slip knot will pass thru the second, third, fourth fixation devices, or any required quantity of fixation devices. The suture will then pass completely through the delivery device where it can be coiled and placed onto a card for storage. Details of illustrative embodiments are discussed below.

(28) To deploy the fixation devices, a series of internal tubes are required. A needle tip would first pierce a tissue and the distal fixation device would be pushed out of the delivery needle via a first tube. This tube will then lock in place inside the handle mechanism and will have further movement and used to house the remaining fixation devices. The delivery needle is then removed from the tissue and reapproximated to pierce the tissue in another location, typically about 5 mm to about 20 mm away. Then the next fixation device in line will be pushed out of the delivery needle. The needle tip is removed and the process will repeat for as many fixation devices as are loaded into the fixed tube.

(29) When all the fixation devices have been placed in the tissue, the suture outside of the of delivery handle is pulled taut which creates a purse string effect. Now the delivery device can be removed and a locking device (e.g., locking mechanism) such as a cinch or special knot can be pushed down to complete and lock the purse string effect. Additional locking mechanisms that may be used include a tag, a barb, a cinch, a toggled needle, or distal knot. The remaining suture is cut.

(30) Alternately an additional fixation device, as described above can be added to the delivery instrument such as to act as a locking device.

(31) The fixation devices are advanced or driven by a handle assembly that employs an internal metering function of the devices for the predetermined length of each fixation device. The handle assembly controls the delivery tubes and locks when required.

(32) In some embodiments, the fixation devices are a string of several hollow braided suture segments placed end to end over a flexible strand (e.g., monofilament, or a braided suture). The string of braided suture segments are stored in a tube and extruded out of the needle for delivery. The string of braided suture segments on the flexible strand (e.g., suture) has a certain column strength that is determined by a given composition and characteristics of the braided suture segments and the flexible strand. The column strength of a given string of braided suture segments may be strengthened (e.g., stiffened) by treating the braided suture segments. Given braided suture segments may be strengthened by heating the braided suture segments. Given braided suture segments may also be strengthened by coating them with a cyanoacrylate coating.

(33) The braided suture segments may be formed of soft materials such as yarns, fibers, filaments, strings, fibrils, strands, sutures, etc., or any combination of such materials. The soft materials may be synthetic or natural materials, or combinations of synthetic and natural materials, and may be bio-degradable or non-degradable within the scope of this disclosure. In one non-limiting embodiment, the soft anchor assembly **10** is made exclusively of soft, suture-based materials.

(34) FIG. 1A and FIG. 1B show schematic plan views of a flexible braided anchor assembly **5** for tissue fixation according to one embodiment of the present disclosure before and after cinching the suture tight. A suture (e.g., flexible strand) connecting the five anchor segments is shown on the

proximal side (above the plane of the page) of the tissue. The five braided anchor segments are positioned on the distal side (below the plane of the page) of a tissue. The anchors are shown in dashed lines since the drawings are intended to show the anchors below the tissue surface.

(35) In FIG. 1A, the flexible braided anchor assembly 5 includes anchors on one side (distal side) of a tissue that is to be repaired, and they are strung together on a suture that is displayed on the opposite side (proximal side) of the tissue. As shown in this embodiment, the five anchors are situated around an opening in the tissue that is to be repaired. The suture is anchored at the first braided anchor segment and connects anchor segments 2-5 with a single suture. The suture is free to extend beyond a final anchor.

(36) In FIG. 1B, the flexible braided anchor assembly 5 has been drawn together to close and fixate the opening by pulling the suture taut. By pulling the suture taut, the exposed edges of the opening in the tissue are drawn together and the opening is fixated (e.g., joined). The opening has been closed by pulling the suture tight. A locking mechanism is engaged with the suture to lock the suture and prevent the fixated tissue from pulling apart. By tightening the suture to draw in the anchors and close the opening in the tissue, the flexible assembly of five braided anchor segments 5 has closed the opening like pulling purse strings to close a purse.

(37) FIGS. 2A-2H show schematic cross section illustrations of some embodiments of delivery devices deploying an assembly of braided anchor segments.

(38) FIG. 2A shows a schematic cross section illustration of a delivery device 15 with a needle 10 having a needle tip 11 after puncturing tissue 100 according to an embodiment of the present disclosure. The needle 10 punctures a first hole (not shown) in the tissue 100. The needle 10 punctures the tissue from a proximal side of the tissue 100 to a distal side of the tissue 100. The needle 10 is at a distal end of an outer tube 90, and the outer tube 90 is coupled to a handle assembly 200.

(39) In some embodiments, the needle 10 and the outer tube 90 may be fabricated as a single piece, whereas in other embodiments, the needle 10 and the outer tube 90 may be assembled from separate pieces. The needle 10 and the outer tube 90 may be fabricated from one or more rigid materials, such as a steel, aluminum, titanium, and so on. The rigid material may also include metallic alloys such as nitinol. In embodiments, when the needle 10 and the outer tube 90 are assembled from separate pieces, the needle 10 and the outer tube 90 may be fabricated from different rigid materials.

(40) FIG. 2B shows a schematic cross section illustration of a first braided suture segment 30 after being extruded from the needle 10 on the distal side of the tissue 100. The first braided suture segment 30 is pushout distally out of the needle by a first drive tube 60. A flexible strand (e.g., suture) 20 is shown threaded through the first braided suture segment 30, looped around, and knotted with a slidable slip knot 22 around the suture 20 between first braided suture segment 30 and the first drive tube 60.

(41) In embodiments, the flexible strand (e.g., suture) 20 may be a monofilament. In other embodiments, the flexible strand 20 may be a suture. The suture 20 may be a braided polyester suture. In embodiments, the suture 20 may be a FiberWire®, TigerWire®, or FiberChain® suture, although any type of suture 20 may be utilized, including cored or coreless sutures. In another embodiment, the flexible strand 20 may be suture tape, such as FiberTape®. The flexible strand 20 may include any soft, flexible strand of material.

(42) In some embodiments, the braided suture segments 30 may be strengthened by heating the braided suture segments 30. The braided suture segments may also be strengthened by coating them with a cyanoacrylate coating.

(43) FIG. 2C shows a schematic cross section illustration of a first braided anchor segment 31 after the suture 20 has been pulled taut a first time. As a result of the suture 20 being pulled tight, the first braided suture segment 30 has been deformed into a first braided anchor segment 31. The first braided suture segment 30 serves as an anchor to the flexible braided anchor assembly 5. By

pulling the suture **20**, the slip knot **22** is tightened to secure the first braided anchor segment on the distal surface of the tissue. The first braided anchor **31** may also be secured by locking the suture in place with any one of locking mechanisms including a tag, a barb, a cinch, a toggled needle, or a distal knot.

(44) In embodiments where the braided suture segments have been strengthened by heating, or by coating them with a cyanoacrylate coating, deforming the braided suture segment by pulling suture **20** cracks or breaks up the coating on the braided suture segment **30**.

(45) FIG. 2D shows a schematic cross section illustration of the needle **10** being repositioned above the tissue **100** prior to puncturing the tissue in a second location. In embodiments, the second location may be between about 3 mm and about 20 mm from the first location, whereas in other embodiments the second location may be between about 5 mm and about 15 mm from the second location, whereas in other embodiments the second location may be between about 8 mm and about 12 mm from the first location. After withdrawing the needle **10** from the tissue **100**, there is approximately 4 cm of suture slack present prior to puncturing the tissue at the second location.

(46) FIG. 2E shows a schematic cross section illustration of a second braided suture segment **40** partially extruded from the needle **10**. The second braided suture segment **40** is pushed from the needle by a second drive tube **70**. The suture **20** passes from the first anchor segment **31** through the first hole (not shown), is strung along the proximal side of the tissue and is fed up through a second hole (not shown) made from puncturing the tissue a second time with the needle **10**. The suture **20** loops around the second braided suture segment **40** and passes through a bore hole (not shown) in the second braided suture segment. The suture is threaded through the braided suture segments remaining in the outer tube **90**.

(47) FIG. 2F shows a schematic cross section illustration of a second braided anchor segment **41** after the suture **20** has been pulled taut a second time. As a result of the suture **20** being pulled taut, the second braided suture segment **40** has been deformed into a second braided anchor segment **41**. The second braided anchor segment **41** is still adjacent to the first drive tube **60** after the suture **20** has been pulled taut a first time.

(48) At this point in the deployment of the flexible braided anchor segment assembly **5**, the suture **20** has been pulled taut twice. First, it was pulled taut to deform the first braided suture segment **30**. Then, the suture was pulled taut a second time to deform the second braided suture segment **40**.

(49) FIG. 2G shows a schematic cross section illustration of the first braided anchor segment **31** and second braided anchor segment **41** after the needle **10** has been withdrawn from the tissue **100** and suture **20** has been pulled taut a third time. Braided anchor segments **31**, **41** are positioned on a distal surface of a tissue **100**. By pulling the suture **20** taut a third time after the needle has been removed, the third pulling of the suture **20** moves the first anchor segment **31** and the second braided anchor segment **41** together. By pulling the first two anchor segments **31**, **41** together, a fold **110** is formed in the tissue **100** between the first two anchor segments **31**, **41**.

(50) The portion of the flexible strand **20** that connects braided anchor segments **31** and **41** on the proximal surface of the tissue **100** has been pulled taut thereby causing the tissue **100** between the braided anchor segments **31** and **41** to be pulled together and to form a fold **110**. The first braided anchor segment has a sliding slipknot **22** that anchors the anchor in the tissue **100**. This figure illustrates braided anchor segments **31**, **41** are positioned on a distal surface of a tissue **100**. The braided anchor segments are interconnected via a flexible strand (e.g., a suture) **20**.

(51) FIG. 2H shows a schematic cross section illustration of the first braided anchor segment **31**, second braided anchor segment **41**, and a third braided anchor segment **51**. The third braided anchor segment **51** is in a stage of being implanted in the tissue **100**. The third braided anchor segment **51** is at the distal end of the delivery needle **10** and is adjacent the second drive tube **60**. In FIG. 2H, the suture has been pulled taut a fourth time deforming the third braided suture segment into the third braided anchor segment **51**.

(52) FIG. 3 is a cross section of an embodiment of a portion of a delivery device including an outer

tube **90** having a delivery needle **10** with a needle tip **11**. The outer tube **90** is preloaded with three braided suture segments **30,40,50** that are interconnected via a suture **20**. The braided suture segments (e.g., segments, or suture segments) are a hollow material which enables suture **20** to pass and slide through each of the braided suture segments. The suture **20** incorporates a sliding slip knot **22** that is placed proximally to segment **30**. Segment **30** is positioned to be the first suture segment to be delivered. As such, it is a primary braided suture segment, because the sliding slip knot is placed proximally to the braided suture segment **30** to allow it to participate in anchoring the primary suture segment **30** on the distal side of the tissue.

(53) A needle tip **11** on a delivery needle **10** is used to pierce (e.g., puncture) the tissue **100** for placement of each segment. The needle tip **11** is a sharp point on the delivery needle **10**. First drive tube **60** is advanced to push primary suture segment **30** out of the needle **10**, and then remains in the advanced position. Suture end (e.g., suture tail) **23** is the cut end of the suture **20** after the formation of the sliding knot **22**. Suture tail **23** may be used as a barb to increase holding strength or aid in preventing pullout of slip knot **22** or first braided anchor segment **31**. Braided suture segment **40** is positioned in first drive tube **60**.

(54) FIG. **4** is a cross section of an embodiment of a portion of a delivery device that has punctured the tissue **100** and has delivered the first braided suture segment **30** to a distal side of tissue **100**. FIG. **4** shows needle **10** and needle tip **11** punctured through tissue **100** with segment **30** pushed out of needle **10** by first drive tube **60**. After extruding segment **30** from the delivery needle **10**, first drive tube **60** remains locked in the position with respect to needle tip **11**. Suture segments **40** and **50** are positioned in first drive tube **60**. In some embodiments, suture segments are pressure fit in first drive tube **60**.

(55) Second drive tube **70** is located in first drive tube **60** with the distal end of second drive tube **70** adjacent the proximal end to braided suture segment **50**. Second drive tube **70** has an outer diameter that is smaller than the inner diameter of first drive tube **60**, and second drive tube **70** is used to push the second through final suture segments out of the distal end first drive tube **60**. Each braided suture segment, beginning with the second braided suture segment is extruded in order from the first drive tube **60** as the column of suture segments is pushed by second drive tube **70**. The suture **20** is threaded through each suture segment. Once the distal end of each successive suture segment is extruded out of first drive tube **60**, the segments exit delivery needle **10** to the distal side of tissue **100**. The second and follow on segments that are extruded from the distal end of first drive tube **60** move through first drive tube **60** and outer tube **90** only in the distal direction. It is surprising that the disclosed delivery process moves in a distal direction only and does not have to retract to load subsequent anchors, nor are the braided suture segments carried on an external slider device. Furthermore, the flexible strand (e.g., suture) traverse the entire length of the bore of the delivery handle and exits the delivery handle. Additional surprising features include that the braided suture segments slide along the first drive tube **60** and are not on or related to being delivered over a rod, and that the delivery device can deploy many devices, from 1 braided suture segment up to and including 25 braided suture segments.

(56) FIG. **5A** shows an embodiment of a braided suture segment **30, 40, or 50** and is hollow allowing for suture **20** to pass and slide through it. In embodiments, there may be from 1 to 25 braided suture segments on a suture **20**, while in some embodiments there may be 1 to 10 braided suture segments on a suture **20**, while in still other embodiments there may be 1 to 5 braided suture segments on a suture **20**, while in still other embodiments there may be 1 to 3 braided suture segments on a suture **20**, while in still other embodiments there may be only 1 braided suture segment on a suture **20**.

(57) FIG. **5B** shows an embodiment of a braided suture segment **30, 40, or 50** that has been modified such that the suture **20** is passed thru the braided fibers of the side wall of the segments **30, 40, 50**, into the cannulation and back out thru the braided fiber side wall again. A feature of this version is that when the segment is placed distally past the tissue and pulled taut, the free ends **37**

and **38** would aid in preventing any pullout back thru the hole created by the needle **10**.

(58) FIG. 5C shows an embodiment of a braided suture segment **80** having a slit or cut **81**. The slit or cut **81** can be formed by a razor blade or any suitable cutting device and its function is to spread the cylindrical device yet further when the suture **20** is pulled taut. The suture **200** would enter the slit **81** causing the slit **81** to split apart or widen. Slits **81** can be on one or both ends of the braided suture segment.

(59) FIG. 6A shows a schematic illustration of an embodiment of braided suture segments on a suture **20**. In this embodiment, a series of suture segments are strung on a suture with a slip knot **22** being positioned after the final braided suture segment on the suture **20**. FIG. 6A shows a first braided suture segment **130**, a second braided suture segment **140**, a third braided suture segment **150**, and a fourth braided suture segment **160** on a suture **20**. In this embodiment, a slip knot **22** is positioned proximal to the fourth braided suture segment **160**. That is, in this embodiment, there is no slip knot present at the proximal end of the first braided anchor segment **31**, but the slip knot is positioned following the final braided suture segment threaded onto the suture.

(60) In other embodiments, a slip knot or other locking mechanism may be positioned proximally to any segments of an assembly of braided suture segments. That is, a slip knot **22** or other locking mechanism may be positioned proximal to any of suture segments **130**, **140**, **150**, or **160**, for example.

(61) FIG. 6B shows a schematic illustration of an assembly of braided anchor segments as they are deployed on the distal side of a tissue. In this embodiment, the assembly of braided suture segments is only anchored to the tissue at where the slip knot is pulled taut. In this embodiment, the assembly of braided anchor segments has a large surface area that enables a large securing surface with only one hole formed in the tissue. This may be useful in retracting tissue from other tissue and holding it in place while other procedures are being undertaken.

(62) FIG. 7A shows a schematic illustration of an embodiment of a long braided suture segment **170** on a suture **20**. In this embodiment the long braided suture segment **170** may be between 40 mm and 70 mm. In this embodiment, a slip knot **22** is positioned proximal to the long braided suture segment **170**. In this embodiment, the long braided anchor **170** segment has a large surface area that enables a large securing surface with only one hole formed in the tissue. This may be useful in retracting tissue from other tissue and holding it in place while other procedures are being undertaken.

(63) FIG. 7B shows a schematic illustration of a long braided anchor segment **171** as it is deployed on the distal side of a tissue. In this embodiment, the long braided anchor segment **171** is only anchored to the tissue at where the slip knot is pulled taut. In this embodiment, the long braided anchor segment **171** has a large surface area that enables a large securing surface with only one hole formed in the tissue. This may be useful in retracting tissue from other tissue and holding it in place while other procedures are being undertaken.

(64) FIG. 8 shows a schematic cross section illustration of a delivery device with two braided anchor segments **61,71** after the suture **20** has been pulled taut a first time. As a result of the suture **20** being pulled tight, the first braided anchor segment **61** and the second braided anchor segment **71** have been deformed into a dual braided anchor segment **61,71**. In this embodiment, the dual braided anchor segment **61,71** has a large surface area that enables a large securing surface with only one hole formed in the tissue. This may be useful in retracting tissue from other tissue and holding it in place while other procedures are being undertaken.

(65) Note that in this embodiment, there is no slip knot present at the proximal end of the first braided anchor segment **61**. Here, the slip knot **22** is tied off proximal to braided suture segment **71**.

(66) FIG. 9A shows an embodiment of a delivery device with a handle assembly **200** with handle cover **210** screwed to handle base **220**. Outer tube **90** is attached to handle base **220** with wheel **230** inside of the handle assembly **200**. Rack **240** includes a visual numerical character to show the position of each braided suture segment as each braided suture segment is deployed. A needle is not

shown on the outer tube **90**.

(67) FIG. **9B** shows an embodiment of a delivery device with a handle assembly **200** with the handle cover **210** removed for visualization. Wheel **230** has an internal drive gear, not shown, that meshes with rack **240**. As wheel **230** is rotated counterclockwise pawl **250** engages in the wheel cogs **231** to stop at predetermined locations. As rack **240** advances, the first drive tube **60** is advanced to extrude a first braided suture segment (e.g., segment) out of the needle (not shown). A spring (not shown) may be included in the handle assembly to drive first drive tube **60**. Once the first braided suture segment is out of the needle, the first drive tube **60** is temporarily fixed to stop **260** until stop **260** engages with handle edge **221**. In embodiments, a spring (not shown) may be positioned between stop **260** and handle edge **221**. This will be the first position for the handle assembly. The first segment is also known as the primary segment, since once the primary segment is deployed, the primary segment **30**, and first drive tube **60** will disengage from rack **240** via a dimple or breakaway. First drive tube **60** will now stay in this fixed position against handle stop **221**.

(68) Subsequent deployment of additional braided suture segments will be accomplished with a second drive tube (not shown). The second drive tube is positioned within first drive tube **60** and attached to rack **240**, and will drive through first drive tube **60** as wheel **230** is rotated to a different positional cog **231** with pawl **250**. This can be repeated for additional fixation devices.

(69) FIG. **10** shows a schematic cross section of an embodiment of a delivery device **15**. An outer tube **90** has a delivery needle **10** and a needle tip **11**. A first drive tube **60** is positioned concentrically inside of the outer tube **90**. First drive tube **60** has an outer diameter that is smaller than the inner diameter of outer tube **90**, and first drive tube **60** is used to push first braided suture segment **30** out of the distal end outer tube **90** through the delivery needle **10**. In embodiments, a distal end of the first drive tube **60** may include a detent **94** that may provide friction so that there is a slight resistive force to the extrusion of braided suture segments. Once the first drive tube **60** has extruded the first suture segment **30**, the first drive tube **60** may become temporarily fixed in the outer tube **90**.

(70) Each braided suture segment, beginning with the second braided suture segment **40** is extruded in order from the first drive tube **60** as the column of suture segments is pushed by second drive tube **70**. A second braided segment **40** and a third braided segment **50** are positioned in the first drive tube **60**. In embodiments, the number of braided suture segments positioned in the first drive tube **60** may be between 1 and 10, and in some embodiments, the number of braided suture segments positioned in the first drive tube **60** may be between 2 and 5. A second braided suture segment **40** through a final suture segment may be preloaded in a first drive tube **60** prior to initiation of a tissue anchoring procedure.

(71) A suture (e.g., flexible strand) may be threaded through the braided suture segments and looped around the first suture segment **30** and tied into a slip knot **22** around the suture **20** between the proximal end of the first braided suture segment **30** and the distal end of the first drive tube **60**.

(72) A second drive tube **70** is positioned concentrically in the first drive tube **60** with its distal end proximal to the third braided suture segment **50** (e.g., the last braided suture segment that is preloaded into the first drive tube **60**). Second drive tube **70** has an outer diameter that is smaller than the inner diameter of first drive tube **60**, and second drive tube **70** is used to push the second through final suture segments out of the distal end first drive tube **60**. Each braided suture segment, beginning with the second braided suture segment is extruded in order from the first drive tube **60** as the column of suture segments is pushed distally by second drive tube **70**. In general, the second drive tube **70** is positioned proximal to the last braided suture segment in the first drive tube **60** regardless of the number of braided suture segments preloaded into the first drive tube **60**.

(73) In some embodiments, a delivery device may not include a first drive tube **60** (not shown). In embodiments, the braided suture segments **30,40,50** may be preloaded into an outer tube **90**, and extruded by pushing the braided suture segments **30,40,50** out of outer tube **90** with the second

drive tube **70**. The braided suture segments **30,40,50** may be press fit into the outer tube **90** to provide friction to the sliding of the braided suture segments **30,40,50** against the pressure applied by the second drive tube **70**.

(74) FIG. **11** shows embodiments of an outer tube **90**, a first drive tube **60**, a second drive tube **70**, and an assembly of braided suture segments strung on a flexible filament (e.g., suture) **20**.

(75) In some embodiments, the outer tube **90** is a single tube that includes a base outer tube **95**, and a needle **10** having a needle point **11**. In some embodiments, the single tube of the outer tube **90**, may or may not have a bushing included. In some embodiments, the outer tube **90** may comprise a two-piece tube having a base outer tube **95** coupled with a needle **10**. In some embodiments, the outer tube **90** may comprise a three-piece tube having the base outer tube **95** coupled with a bushing and the needle **10**. In embodiments when the outer tube **90** comprises more than 1 part, the various parts may be fabricated from different rigid materials. In other embodiments when the outer tube **90** comprises more than 1 part, the various parts may be fabricated from similar rigid materials.

(76) FIG. **11** also shows embodiments of the first drive tube **60**. In some embodiments, the first drive tube **60** is a single tube. In some embodiments, the first drive tube **60** may comprise a two-piece tube having a base first drive tube **65**, and a first drive tube tip **62**. In embodiments when the first drive tube **60** comprises more than 1 part, the various parts may be fabricated from different rigid materials. In other embodiments when the first drive tube **60** comprises more than 1 part, the various parts may be fabricated from similar rigid materials.

(77) FIG. **11** further shows embodiments of the second drive tube **70**. The second drive tube **70** is made from rigid materials. The rigid materials may include one or more rigid materials, such as a steel, aluminum, titanium, and so on. The rigid material may also include metallic alloys such as nitinol.

(78) FIG. **11** still further shows embodiments of the an assembly of five braided suture segments **30,40,50,52,54** with a flexible filament (e.g., suture) **20** threaded through bore holes that go through a length-wise dimension of the braided suture segments.

(79) FIG. **12** shows steps of an embodiment of a method for implanting braided anchor segments connected via a single flexible strand (e.g., suture) for tissue fixation.

(80) At **1210**, puncture a tissue at a location with a delivery needle. The needle having a needle tip punctures the tissue from a proximal side of the tissue to a distal side of the tissue. The needle is at a distal end of an outer tube, and the outer tube is coupled to a handle assembly.

(81) At **1220**, extrude a braided suture segment, slidably coupled to a flexible strand, from the delivery needle. The braided suture segment is extruded from the needle on the distal side of the tissue. The braided suture segment is pushout distally out of the needle by a first drive tube.

(82) At **1230**, pull the flexible strand to deform the braided suture segment forming the braided anchor segment. As a result of the flexible strand being pulled tight, the first braided suture segment is deformed into a first braided anchor segment.

(83) At **1240**, withdraw the delivery needle from the tissue. The delivery needle is withdrawn from the tissue leaving behind the braided anchor segment on the distal side of the tissue.

(84) At **1250**, determine if another anchor is to be deployed. A determination is made as to whether an additional braided anchor segment is required to fixate the tissue. If it is determined that an additional braided anchor segment is necessary to fixate the tissue, then the method returns to step **1210** to initiate placing an additional braided anchor segment on the distal side of the tissue. If it is determined that an additional braided anchor segment is not necessary to fixate the tissue, then the method advances to step **1260** to secure the flexible strand.

(85) At **1260**, secure the flexible strand to keep the tissue fixated by the braided suture segment(s). The flexible strand (e.g., suture) is secured by engaging a locking mechanism (e.g., cinch) with the suture to lock the suture and prevent the fixated tissue from pulling apart. By tightening the suture to draw in the anchors and close the opening in the tissue, the flexible assembly braided anchor

segments has closed the opening like pulling purse strings to close a purse.

(86) At **1270**, cut the suture. The flexible strand is cut on the distal side of the locking mechanism after the flexible strand has been pulled taut and secured by the locking mechanism.

(87) The embodiments of the invention described above are intended to be merely exemplary; numerous variations and modifications will be apparent to those skilled in the art. Such variations and modifications are intended to be within the scope of the present invention as defined by any of the appended claims.

Claims

1. A method for implanting a plurality of braided suture segments connected via a single flexible strand for tissue fixation, comprising; puncturing the tissue at a first location with a delivery needle; extruding a first braided suture segment from the delivery needle, the first braided suture segment is the first of a plurality of braided suture segments slidably coupled to a flexible strand, the flexible strand being threaded through a bore hole in the first braided suture segment; pulling the flexible strand a first time to deform the first braided suture segment forming the first braided anchor segment, pulling the flexible strand the first time tightens a slipknot around a proximal end of the first braided suture segment, tightening the slipknot is configured to anchor the first braided anchor segment to the tissue to prevent the first braided anchor segment from pulling through the tissue; withdrawing the delivery needle from the tissue a first time; puncturing the tissue at a second location with the delivery needle; extruding a second braided suture segment from the delivery needle; pulling the suture a second time to deform the second braided suture segment forming the second braided anchor segment; withdrawing the delivery needle from the tissue a second time; pulling the suture a third time, the pulled suture causing the second anchor segment to be moved toward the first braided anchor segment; securing the suture to keep the tissue fixated by the plurality of braided suture segments; and cutting the suture, the suture is cut a distance away from the proximal end of a final braided anchor segment of the plurality of the braided anchor segments.
2. The method of claim 1, following pulling the suture a third time, and before securing the suture, further comprising: puncturing the tissue at a third location with the delivery needle; extruding a third braided suture segment from the delivery needle; pulling the suture a fourth time to deform the third braided suture segment forming the third braided anchor segment; withdrawing the delivery needle from the tissue a third time; and pulling the suture a fourth time, the pulled suture causing the third braided anchor segment to be moved toward the second braided anchor segment.
3. The method of claim 1, wherein: the first braided suture segment is a primary first braided suture segment; the extruding the first braided suture segment is accomplished by advancing first drive tube to push the first braided suture segment out of the delivery needle; and after advancing first drive tube to push the first braided suture segment out of the delivery needle, first drive tube remains in an advanced position temporarily.
4. The method of claim 1, wherein the extruding a second braided suture segment is accomplished by advancing second drive tube to push the second braided suture segment out of the delivery needle.
5. The method of claim 1, wherein the second location is between about 5 mm and about 15 mm from the first location.
6. The method of claim 1, wherein the securing the suture comprises a locking device or an overhead knot that can be run down to a fixation site.
7. The method of claim 1, wherein the securing the suture comprises locking the suture in place using a locking mechanism comprising at least one of a tag, a barb, a cinch, a toggled needle, or a distel knot.
8. A surgical device for implanting a flexible braided anchor assembly for tissue fixation,

comprising: a handle assembly, the handle assembly having a cylindrical bore passing through the handle assembly on a center axis along a length of the handle assembly; an outer tube positioned in the cylindrical bore of the handle assembly, the outer tube being hollow and having a common center line with the cylindrical bore of the handle assembly, the outer tube comprising a delivery needle having a needle tip at a distal end of the outer tube; a plurality of braided suture segments slidably coupled to a flexible strand, the flexible strand being translationally threaded through plurality of braided suture segments; a first drive tube positioned in the outer tube and having a common center line with the outer tube, the first drive tube configured to extrude a first braided suture segment; and a second drive tube positioned in the first drive tube and having a common center line with the first drive tube, the second drive tube configured to extrude remaining braided suture segments of the plurality of braided suture segments, wherein the remaining braided suture segments of the plurality of braided suture segments are translationally threaded through and pressure fit within the first drive tube.

9. The surgical device of claim 8, wherein each of plurality of braided suture segments comprises a hollow braided suture segment.

10. The surgical device of claim 8, wherein: the suture comprises a monofilament strand; and the plurality of braided suture segments comprises a plurality of hollow braided suture segments aligned end to end having the monofilament strand threaded through the aligned plurality of hollow braided suture segments.

11. The surgical device of claim 8, wherein a column strength of the plurality of hollow braided suture segments is increased by application of at least one of heat and a cyanoacrylate coating to each of the segments of the plurality of hollow braided suture segments.

12. The surgical device of claim 11, wherein a distal end of the flexible strand threaded through the first braided suture segment is looped around the first braided suture segment and knotted around the flexible strand between the first braided suture segment and the first drive tube.

13. The surgical device of claim 8, wherein a braided suture segment of the plurality of braided suture segments has a length of between about 5 mm and about 50 mm.

14. The surgical device of claim 9, wherein each of the braided suture segments of the plurality of braided suture segments has a length of between about 5 mm and about 12 mm.

15. The surgical device of claim 8, wherein each of the plurality of braided suture segments is treated to increase a rigidity of the plurality of braided suture segments slidably coupled to the flexible strand.

16. The surgical device of claim 15, wherein the treatment comprises: heating each of the plurality of braided suture segments; or applying a cyanoacrylate coating to each of the plurality of braided suture segments.
